

Artificial Intelligence

Assignment#03



23K-2001

BCS-6J

Student Information

Name: Muzammil Siddiqui

Roll No: 23K-2001

Section: BCS-6J

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1. Competition Overview

- **Competition Name:** Playground Series S6E4
- **Platform:** Kaggle
- **Problem Type:** Multiclass Classification
- **Target Variable:** Irrigation_Need
- **Evaluation Metric:** Accuracy
- **Objective:** Predict irrigation requirement category for each record in the test dataset.

Brief Understanding of Dataset

The dataset contains agricultural/environmental features related to irrigation conditions. Based on these input variables, the model predicts one of multiple irrigation need classes.

Files provided by competition:

- train.csv → training data with labels

- test.csv → unseen test data without labels
- sample_submission.csv → required output format

2. Data Preprocessing

Missing Values Handling

The dataset was checked for missing values. No major missing-value issue affected training.

Encoding Techniques

Categorical/text columns were converted into numeric format using **Label Encoding** so machine learning algorithms could process them.

Feature Scaling

Feature scaling was not required for tree-based models such as Decision Tree and CatBoost.

Train-Validation Split

The dataset was divided into:

- **80% Training Data**
- **20% Validation Data**

Stratified splitting was used to preserve class balance.

3. Models Attempted

The following machine learning models were trained and evaluated:

Model	Validation Accuracy
Decision Tree	0.9843
Naive Bayes	0.7445
CatBoost Classifier	0.9853

Model Notes

Decision Tree

Performed very strongly with high accuracy. Fast to train and easy to interpret.

Naive Bayes

Low accuracy because it assumes feature independence, which was not suitable for this dataset.

CatBoost

Best performing model. It handled feature relationships efficiently and generalized better than other models.

4. Cross Validation Strategy

Due to large dataset size, an **80/20 train-validation split** was used instead of LOOCV (Leave-One-Out Cross Validation), which would be computationally expensive.

K-Fold concepts were considered, but a single validation split was selected for efficiency because of assignment deadline and dataset size.

Best Validation Accuracy

0.9853 (CatBoost)

5. Failed Attempts and Insights

Attempt 1 – Naive Bayes

- **Accuracy:** 0.7445
- **Issue:** Model assumptions were too simple.
- **Insight:** Features were not independent, reducing performance.

Attempt 2 – Decision Tree

- **Accuracy:** 0.9843
- **Issue:** Slightly lower than CatBoost.
- **Insight:** Strong baseline model, but ensemble boosting improved results further.

6. Final Model Selection

Best Model

CatBoostClassifier

Hyperparameters Used

iterations = 300

depth = 8

learning_rate = 0.08

loss_function = 'MultiClass'

Why Selected?

- Highest validation accuracy
- Strong handling of structured/tabular data
- Robust multiclass classification performance
- Better generalization than Decision Tree

7. Leaderboard Performance

- **Kaggle Score:** 0.95840
- **Rank:** 2940

2937	Gaurav Rawat		0.95842	3	7d
2938	DannyDog1227		0.95842	6	5d
2939	GCarnovale		0.95841	1	4d
2940	Muzammil Siddiqui		0.95840	1	29m
 Your First Entry! Welcome to the leaderboard!					
2941	Frank Ruta		0.95840	3	24d
2942	SK0101		0.95839	2	23d

8. Conclusion and Learnings

This assignment provided practical experience with a real-world machine learning competition on Kaggle.

Key Learnings

- Importance of comparing multiple models
- Proper preprocessing of categorical data
- Validation before final submission
- Tree-based boosting methods perform strongly on tabular datasets

Challenges Faced

- Large dataset size increased training time

- Limited Kaggle submissions required careful model selection

Future Improvements

- Hyperparameter tuning using Grid Search
- K-Fold Cross Validation
- Ensemble stacking methods
- Feature engineering for further accuracy gains